

Education

University of North Carolina at Chapel Hill	August 2024 - May 2026
Master of Science in Computer Science	GPA 4.0 / 4.0
Indian Institute of Technology Bombay	August 2019 - May 2023
Bachelor of Technology in Computer Science and Engineering	GPA 8.6/10.0

Technical Skills

- **Languages:** C++, C#, Python, TypeScript/JavaScript, SQL
- **Systems & Tools:** Unity, Apple Metal (GPU compute), FastAPI, React, CMake, Git, multi-threading, real-time scheduling
- **ML & Data:** PyTorch, LightGBM, HuggingFace, pandas, NumPy

Professional and Research Experience

Project Research Assistant	Mumbai, India
Koita Centre for Digital Health	September 2023 - June 2024
○ Implemented a novel data augmentation method using segmented KNN imputation on multiomics data to add missing values	
○ Developed a novel VAE -based sparse branching architecture to generate multimodal data, achieving test accuracy of 93%	

Data-efficient machine learning for chest x-ray classification	IIT Bombay
Assistant Prof. Kshitij Jadhav	January 2023 - May 2023
○ Optimized the Spear-DECILE library for medical image analysis using labeling function techniques and domain knowledge	
○ Trained a CNN with the optimized library at 0.82 precision and 0.86 recall, with a 22% training speedup	

Engineering Intern	Pune, India
FinIQ Consulting India Pvt Ltd	May 2022 - July 2022
○ Developed a responsive, dynamic Client Holdings Portfolio Dashboard using the Angular web framework with API integration	
○ Integrated keyboard event handling and optimized a Foreign Exchange (FX) Options prototype using multi-threading	
○ Optimized backend operations for FX Options through data parallelization, achieving a 2.1x reduction in server-to-client latency	

Key Projects

Adaptive XR Hand-Interaction Governor	UNC Chapel Hill
Unity, C#, Python, LightGBM	Spring 2026
○ Engineered a real-time three-state runtime governor (Idle→Approach→Active) in Unity/C# that downshifts hand-interaction support work on a Meta Quest 3 , with dwell and hysteresis guards to prevent state oscillation	
○ Deployed a flattened C# decision-tree policy distilled from a trained LightGBM model (95.19% accuracy), cutting app GPU time 8.6% and raising task completions 11.2% versus a static high-support configuration	

stlviz — C++ STL Visualization Library	UNC Chapel Hill
C++, SFML, CMake, MVC	Fall 2025
○ Built a C++ library that visualizes STL containers and operations at runtime via drop-in <code>vstd</code> wrapper classes, letting a user's own code drive a live SFML visualizer	
○ Structured it as an MVC, cross-platform (Linux/Windows) CMake package; an HCI user study rated it 6.3 for debugging effectiveness and 5.7 for UI intuitiveness on a 7-point Likert scale	

Real-Time Task-Based Boids Simulation Engine	UNC Chapel Hill
C++, Apple Metal, RM/EDF scheduling	Fall 2025
○ Refactored a monolithic simulation into a schedulable real-time task set with RM/EDF scheduling, spatial hashing ($O(n \cdot k)$), and overload management with graceful degradation in C++	
○ Added multiprocessor EDF and an Apple Metal GPU compute layer: 7.9x multithread speedup, 5.4x GPU speedup, and schedulable population raised from 3,000 to 10,000 boids	

Unreliable Narrator Analysis Service	UNC Chapel Hill
Python, FastAPI, Ollama, Pydantic	Spring 2025
○ Built a full-stack LLM service with a FastAPI backend serving local Llama/Mistral models via Ollama , with Pydantic validation and JSON persistence	
○ Classified unreliable-narrator types and returned highlighted spans with per-type explanations, delivered with a UI	

Brain-to-Text BCI Decoder Efficiency	UNC Chapel Hill
PyTorch, Mamba/SSM, quantization, pruning	Fall 2025
○ Implemented GRU and Mamba state-space phoneme decoders in PyTorch and benchmarked phoneme error rate, latency (p50/p95/p99), memory, and energy under FP32/FP16/BF16 inference and four pruning methods	
○ Shipped a reproducible pipeline — setup, data download, training, and evaluation scripts — deployed on GPU/HPC infrastructure (Vast.ai, Longleaf)	